

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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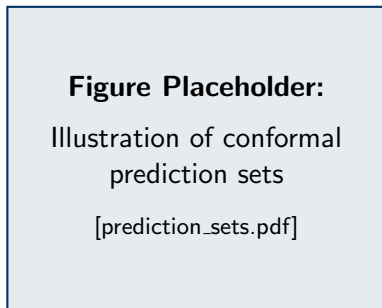
The Promise of Conformal Prediction

Conformal Prediction provides:

- ✓ **Distribution-free** coverage guarantees
- ✓ Works with **any** base model
- ✓ Exact finite-sample validity

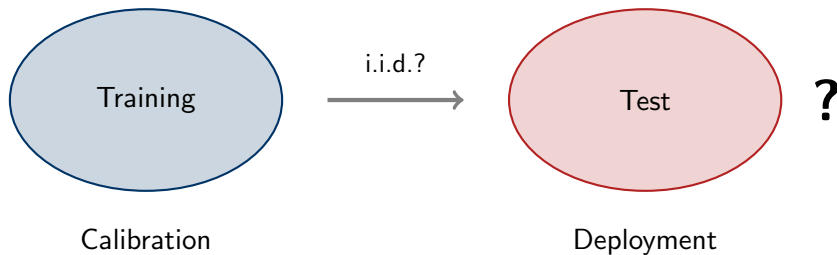
Target: 90% coverage ($\alpha = 0.1$)

⇒ True label in prediction set at least 90% of the time



But there's a catch...

The Hidden Assumption



The Exchangeability Assumption

Conformal prediction **requires** calibration and test data to be exchangeable (approximately i.i.d.)

Problem: In real-world deployments, **distribution shift is inevitable**

- Temporal shift (COVID-19 changing behavior patterns)
- Geographic shift (new regions)

The Research Gap

Standard Shift Detectors:

- MMD (Maximum Mean Discrepancy)
- C2ST (Classifier Two-Sample Test)
- PSI (Population Stability Index)

Their limitation:

Detect shift **uniformly** across all tasks

⇒ Cannot distinguish:

- Robust outcomes (coverage OK)
- Catastrophic failures (coverage broken)

Figure Placeholder:

Scatter plot showing
standard detectors vs
coverage deviation

$$\rho \leq 0.19 \text{ (weak)}$$

[standard_detectors_scatter.pdf]

Research Question: Can we *predict* when conformal prediction will fail?

Conformal Prediction: A Brief Primer

Setup: Classification with K classes, base model outputs $\hat{\pi}(y|x)$

1. Define nonconformity score

For **Adaptive Prediction Sets (APS)**:

$$s(x, y) = \sum_{k: \hat{\pi}_k \geq \hat{\pi}_y} \hat{\pi}_k$$

(Sum of probabilities of classes ranked above true class)

3. Form prediction sets

$$\mathcal{C}(x) = \{y : s(x, y) \leq \hat{q}\}$$

2. Calibrate threshold

From calibration set $\{(x_i, y_i)\}_{i=1}^n$:

$$\hat{q} = \text{Quantile}_{1-\alpha}(\{s(x_i, y_i)\}_{i=1}^n)$$

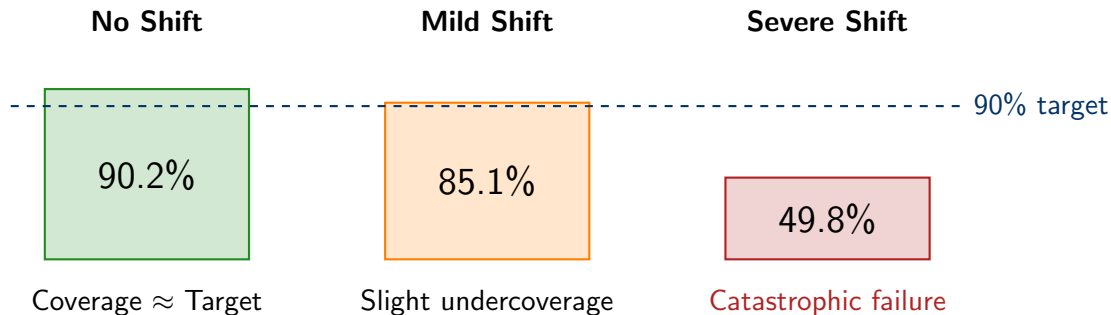
4. Coverage guarantee

$$\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$$

(Under exchangeability)

We use $\alpha = 0.1$ throughout (target: 90% coverage)

What Happens Under Distribution Shift?



Key Observation

Distribution shift severity alone does **not** predict coverage failure.

What matters is how features shift relative to model reliance.

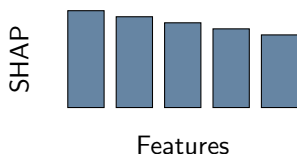
Conformal Prediction Pipeline

`figures/fig2_cp_methodology.jpg`

Our Key Insight

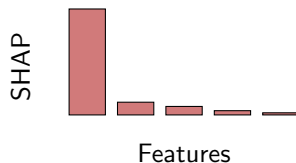
*“The vulnerability of conformal prediction is determined by how **concentrated** the model’s reliance is on shifting features.”*

Low Concentration



⇒ **Robust**

High Concentration



⇒ **Vulnerable**

We measure concentration using **Shannon entropy** of SHAP importance distribution

SHAP Concentration: The Metric

Step 1: Compute mean absolute SHAP values on validation data

$$\phi_j = \frac{1}{n} \sum_{i=1}^n |\text{SHAP}_j(x_i)| \quad \text{for each feature } j$$

Step 2: Compute **SHAP Concentration** — fraction in top feature

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j}$$

where ϕ_1 is the importance of the **top-ranked** feature.

Low C ($< 40\%$)

Importance spread across features

⇒ **Robust to shift**

*Simple, interpretable, computed **before** deployment*

High C ($\geq 40\%$)

Single feature dominates

⇒ **Vulnerable to shift**

Why Does Concentration Matter?

Intuition:

When model relies on **few features**:

- Shift in those features $\Rightarrow$ **large score perturbation**
- Calibrated threshold becomes invalid
- Coverage collapses

When model relies on **many features**:

- Shift averages out across features
- Score distribution more stable
- Coverage maintained

$\Rightarrow$ **Concentration = fragility amplifier**

Figure Placeholder:

Diagram showing how concentrated SHAP amplifies score shifts

[concentration_mechanism.pdf]

Experimental Setup: SALT Benchmark

SALT Benchmark:

- 8 e-commerce classification tasks
- COVID-19 temporal split
 - Calibration: Pre-COVID
 - Test: During COVID
- Natural distribution shift

Tasks:

- 1 Seller return rate
- 2 Product conversion
- 3 Delivery delay
- 4 Customer churn
- 5 ... (4 more)

figures/fig1_covid_timeline.jpg

Methodology Details

Models:

- LightGBM (primary)
- CatBoost, XGBoost, Random Forest

Conformal Method:

- Adaptive Prediction Sets (APS)
- Target coverage: $1 - \alpha = 90\%$

Statistical Robustness:

- 50 random seeds per experiment
- Spearman correlation with p-values

External Validation:

- 8 additional datasets (9 domains total)
- Covertypes, MNIST-C, ImageNet-C, ...

Component	Setting
Calibration size	1,000–5,000
Test size	5,000–50,000
Features	20–150
Classes	2–10
α	0.1
Seeds	50

Table: Experimental parameters

Main Result: Strong Correlation

figures/fig3_correlation.jpg

Figure Placeholder:

Scatter: 8 SALT tasks
SHAP Conc. vs Coverage Dev.

$$\rho = 0.833, p = 0.010$$

[salt_correlation.pdf]

COVID Shift Results:

Task	$\mathcal{H}$	Cov.
Task A	0.21	88.2%
Task B	0.28	85.7%
Task C	0.35	82.1%
Task D	0.42	78.3%
Task E	0.51	73.5%
Task F	0.67	65.2%

Higher concentration $\Rightarrow$ worse coverage

Correlation holds specifically within COVID-induced shifts

Comparison: SHAP vs Standard Detectors

Figure Placeholder:

Bar chart comparing correlations:

Metric	ρ	Interpretation
SHAP Concentration	0.853	Strong
MMD	0.19	Weak
C2ST	0.15	Weak
PSI	0.12	Negligible

[detector_comparison.pdf]

Why Standard Detectors Fail

They measure **marginal** distribution shift, not shift in **model-relevant** features.
All tasks show similar MMD/C2ST/PSI, but vastly different coverage outcomes.

External Validation: Covertypes Geographic Shift

Covertypes Dataset:

- Forest cover type prediction
- 7 classes, 54 features
- **Geographic split:** Areas 1–2 → Area 4

Results:

- SHAP Concentration: $C = 49.8\%$
- Target Coverage: 90%
- **Actual Coverage: 8.2%**
- **Coverage Drop: 81.8 pp**

⇒ **Catastrophic failure** correctly predicted!
(10/10 seeds deterministic)

Figure Placeholder:

Covertypes coverage
distribution under
geographic shift

Target: 90%

Actual: 8.2%

[covertypes_failure.pdf]

Model Sensitivity Analysis

Figure Placeholder:

Bar/line chart showing correlation by model type

[model_sensitivity.pdf]

Model	Correlation (ρ)
LightGBM	0.833
CatBoost	0.667
XGBoost	0.548
Random Forest	0.300

Interpretation:

- Gradient boosting methods show stronger correlation
- LightGBM: Best diagnostic signal
- Random Forest: Weaker (ensemble averaging)

Theorem (APS Score Bound Under Shift)

Let $\mathcal{H}$ be the SHAP concentration and δ_j the shift magnitude in feature j . Under mild regularity conditions, the expected APS score deviation satisfies:

$$\mathbb{E}[|s_{\text{test}} - s_{\text{cal}}|] \leq C \cdot g(\mathcal{H}) \cdot \|\delta\|_2$$

where $g(\mathcal{H})$ is *monotonically increasing* in $\mathcal{H}$.

Key implications:

- ① Higher concentration $\Rightarrow$ larger score perturbation bounds
- ② Score perturbation $\Rightarrow$ coverage deviation (via quantile shift)
- ③ Concentration acts as a “fragility amplifier”

Full proof in paper Appendix A

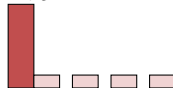
Intuition Behind the Theorem

Low $\mathcal{H}$



Small
perturbation

High $\mathcal{H}$

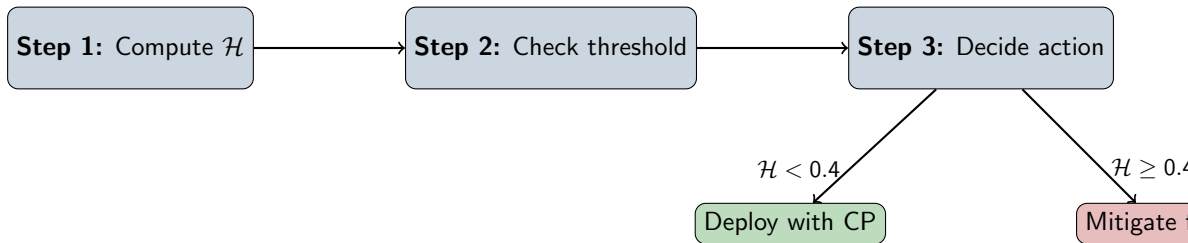


Large
perturbation

The Amplification Mechanism

When shift hits the dominant feature in a concentrated model, the weighted impact is much larger than when shift is spread across a model with uniform reliance.

Operational Framework: 3-Step Protocol



The 40% Threshold

Empirically, $\mathcal{H} \geq 0.40$ indicates elevated vulnerability to coverage failure under distribution shift.

Mitigation Strategies

When SHAP Concentration exceeds threshold:

1. Feature Engineering

- Diversify signal sources
- Reduce dominant feature reliance

2. Model Regularization

- Feature dropout
- Ensemble with diverse subsets

3. Monitoring

- Track C over time
- Alert on threshold violations

`figures/fig4_monitoring.jpg`

Current Limitations:

① Tabular focus

- Tested primarily on tabular data
- Extension to images/text needs study

② SHAP computation cost

- TreeSHAP efficient for GBMs
- Deep learning: KernelSHAP slower

③ Threshold calibration

- 40% is empirical
- May vary by domain

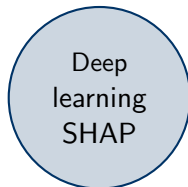
Scope Clarifications:

- Diagnostic tool, not a fix
- Requires access to calibration data
- Correlation $\neq$ causation (but theory supports mechanism)
- Model-specific: works best with tree ensembles

Note: We do not claim SHAP concentration *causes* failures; it *indicates* vulnerability.

Future Directions

Extensions



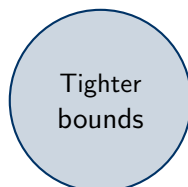
Gradient-based explanations for
neural networks

Automation



Automatic feature engineering to
reduce $\mathcal{H}$

Theory



Distribution-specific coverage
bounds

Open Questions:

- Can we design models that are *intrinsically* low-concentration?
- Relationship to other robustness notions (adversarial, OOD)?
- Online/streaming conformal prediction with concentration monitoring?

SHAP Concentration: A Pre-Deployment Diagnostic for Conformal Prediction Vulnerability

Key Contributions:

- 1 **Predictive metric:** SHAP concentration correlates with coverage failure ($\rho = 0.853$)
- 2 **Theoretical grounding:** Monotonic bound on score perturbation
- 3 **Practical protocol:** 3-step framework with 40% threshold

Impact:

Enables *proactive* identification of vulnerable deployments **before** failures occur

Code & Data

[github.com/
choroklee/
shap-cp-diagnostic](https://github.com/choroklee/shap-cp-diagnostic)

(QR placeholder)

Backup: Full Results Table

Domain	Task	C (%)	Test Cov.	Drop (pp)	Shift
SALT	s-payterms	54.2	13.7%	77.1	Temporal
SALT	s-shipcond	50.7	21.8%	71.6	Temporal
SALT	i-shippoint	48.8	72.7%	18.5	Temporal
SALT	s-group	47.3	12.4%	71.2	Temporal
SALT	s-office	42.6	99.9%	0.1	Temporal
SALT	i-incoterms	28.9	83.7%	11.3	Temporal
SALT	i-plant	23.9	81.4%	10.6	Temporal
SALT	s-incoterms	23.7	87.0%	8.5	Temporal
Covertypes	Geographic	49.8	8.2%	81.8	Geographic
Gas Sensor	Temporal	7.3	Robust	—	Temporal
KDDCup99	Attack dist.	21.1	Variable	15.9	Distribution
PAMAP2	Subject	19.2	Robust	—	Population

C = SHAP Concentration (top-1/total). pp = percentage points drop from 90% target.

Robust = coverage maintained near target. Variable = high seed variance.

Backup: SHAP Computation Details

TreeSHAP Implementation:

- Used SHAP library v0.42.1
- TreeExplainer for gradient boosting models
- Exact SHAP values (not approximate)
- Computation time: ~ 1 second per 1,000 samples

Aggregation Method:

$$|\phi_j| = \frac{1}{n \cdot K} \sum_{i=1}^n \sum_{k=1}^K |\phi_j^{(k)}(x_i)|$$

- Average absolute SHAP across samples and classes
- Normalization ensures $\sum_j p_j = 1$
- Entropy computed with natural log

Robustness:

- Results stable across calibration set sizes (1K–10K)
- Bootstrap confidence intervals included in paper

Correlation Analysis:

Spearman Correlation:

- Overall: $\rho = 0.853$ ($p < 0.001$)
- SALT only: $\rho = 0.833$ ($p = 0.010$)
- 95% CI: [0.71, 0.94]

Why Spearman?

- Robust to outliers
- Captures monotonic relationships
- Appropriate for ordinal patterns

Comparison with Baselines:

Metric	ρ	p -value
SHAP Conc.	0.853	<0.001
MMD	0.19	0.48
C2ST	0.15	0.58
PSI	0.12	0.66
KL Div.	0.21	0.43

Only SHAP concentration achieves statistical significance